Preprocessing

11/1/2023

Description

This file enables preprocessing of paleolithic sign sequences from the Swabian Aurignacian and the Magdalenian "necklace' ' of the Saint-Germain-la-Rivière burial.

Session Info

Give the session info (reduced).

```
## [1] "R version 4.2.2 Patched (2022-11-10 r83330)"
```

```
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
library(dplyr)
```

Preprocessing

Magdalenian

Preprocessing and storage of cleaned file for signs of the Magdalenian "necklace' '. Load data from github repo and remove missing rows.

```
# load data from github repo
magda.data <- read.csv(
  "~/Github/UpperPalCombinatorics/data/magdalenian/StGermain_original.csv")
# select columns
magda.data <- select(magda.data, c('VanhaerenDErrico2003_Fig.6', 'DutkiewiczBentz2023',
                                   'lateralization','position', 'coding', 'certainty'))
# remove rows for which there is no sign coding
magda.data <- magda.data[!magda.data$coding == "", ]
```

Remove UTF8 characters which do not represent actual signs on the objects, i.e "punctuation".

```
# remove some special characters
coding.clean <- gsub('[!?]', '', magda.data$coding)
# remove sign strings which are not discrete, i.e. overlapping sign types
# -> indicated by parentheses '()'
```

```
coding.clean <- gsub("\\([^()]*\\)", "", coding.clean)
# remove sign strings which are not in linear order, i.e. those grouped in visual fields
# -> indicated by square brackets '[]'
coding.clean <- gsub("\\[(.*?)\\]", "", coding.clean)
# print for visual inspection
# print(coding.clean)

# add column to original data frame again
magda.data$coding_clean <- coding.clean
```

Safe to file.

```
write.csv(magda.data,
          file = "~/Github/UpperPalCombinatorics/data/magdalenian/StGermain_clean.csv",
          row.names = F)
```

Aurignacian

Preprocessing and storage of cleaned file for the signs of the mobile objects from the Swabian Jura. Load data from github repo and remove missing rows.

```
# load data from github repo
aur.data <- read.csv(
  "~/Github/UpperPalCombinatorics/data/aurignacian/signBase_Version1.0_CodingsUTF8.csv")
# select columns
aur.data <- select(aur.data, c('object_id', 'site_name', 'date_bp_max_min',
                                 'material','object_type', 'coding'))
# remove rows for which there is no sign coding
aur.data <- aur.data[!aur.data$coding == "", ]
```

Remove UTF8 characters which do not represent actual signs on the objects, i.e "punctuation".

```
# remove special characters
coding.clean <- gsub('[!?]', '', aur.data$coding)
# remove sign strings which are not discrete, i.e. overlapping sign types
# -> indicated by parentheses '()'
coding.clean <- gsub("\\([^()]*\\)", "", coding.clean)
# remove sign strings which are not in linear order, i.e. those grouped in visual fields
# -> indicated by square brackets '[]'
coding.clean <- gsub("\\[(.*?)\\]", "", coding.clean)
# print for visual inspection
#print(coding.clean)

# add column to original data frame again
aur.data$coding_clean <- coding.clean
```

Safe to file.

```
write.csv(aur.data,
          file = "~/Github/UpperPalCombinatorics/data/aurignacian/swabianJura_clean.csv",
          row.names = F)
```